

# C24 Dynamical Systems

## Lecture 2: Equilibria and Stability

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# Lecture 2: Equilibria and Stability

- Equilibrium definitions
- Stability definitions for equilibrium points
- Phase space of linear systems
- Linearization and the stability of equilibria of nonlinear systems

# Equilibria of continuous time systems

An **equilibrium** is a point in state space where  $\dot{\mathbf{x}} = 0$ :

$\mathbf{x}^*$  is an equilibrium of  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$  if and only if  $\mathbf{f}(\mathbf{x}^*) = 0$

- If  $\mathbf{x}(0) = \mathbf{x}^*$ , then  $\mathbf{x}(t) = \mathbf{x}^*$  for all  $t$   
 $\mathbf{x}^*$  is sometimes called a **fixed point**
- For a **linear** autonomous system with non-zero eigenvalues, there is only one solution to  $A\mathbf{x}^* = 0$ , namely  $\mathbf{x}^* = 0$
- In general there may be many points  $\mathbf{x}^*$  satisfying  $\mathbf{f}(\mathbf{x}^*) = 0$  therefore a **nonlinear** system can have many equilibria

# Equilibria of discrete time maps

**Discrete time** systems also have equilibrium points

$\mathbf{x}^*$  is an equilibrium of  $\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k)$  if and only if  $\mathbf{f}(\mathbf{x}^*) = \mathbf{x}^*$

- The equilibria of a discrete time system are the **fixed points** of  $\mathbf{f}(\mathbf{x})$ , so

$$\mathbf{x}_0 = \mathbf{x}^* \implies \mathbf{x}_k = \mathbf{x}^* \text{ for all } k$$

- For differential equations, there is a flow of solutions through phase space but the state of a discrete time system ‘jumps’ between points space, making their trajectories harder to visualise

# Flows and equilibria

- We can think of the solution trajectories of a nonlinear ODE as a flow in  $n$ -dimensional state space:  
system trajectories are analogous to streamlines in a 3-dimensional fluid

- In this analogy the vector-valued function  $\mathbf{f}$  in

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x})$$

is a vector field defining the **flow velocity**

- Flows can end or begin at equilibria, or circulate around them
- The stability of the flow near an equilibrium is an important characteristic, which we will focus on today

# Stability of flow equilibria

- **Definition:** An equilibrium point  $\mathbf{x}^*$  is said to be **stable** if, given any  $\varepsilon > 0$ , there exists  $\delta > 0$  such that all solutions  $\mathbf{x}(t)$  satisfy

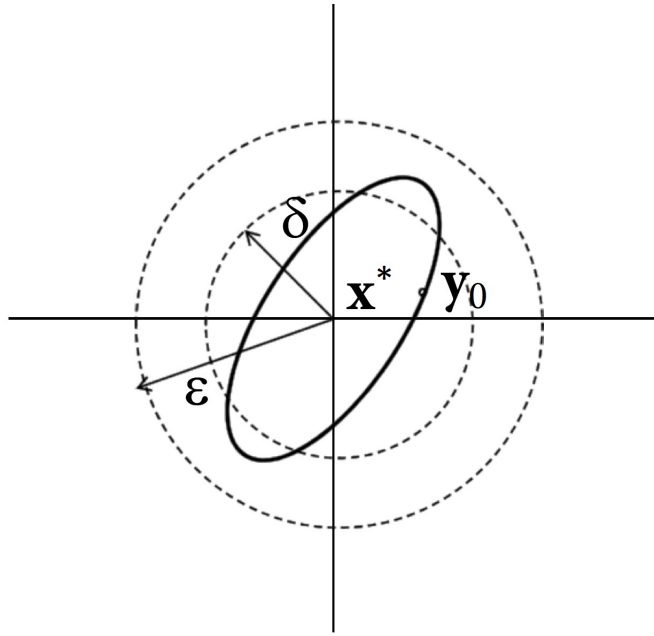
$$\|\mathbf{x}(t) - \mathbf{x}^*\| < \varepsilon \text{ for all } t \geq 0 \text{ whenever } \|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$$

- Otherwise the equilibrium point is said to be **unstable** (i.e. if, for some  $\varepsilon > 0$ , no  $\delta > 0$  exists satisfying this condition)

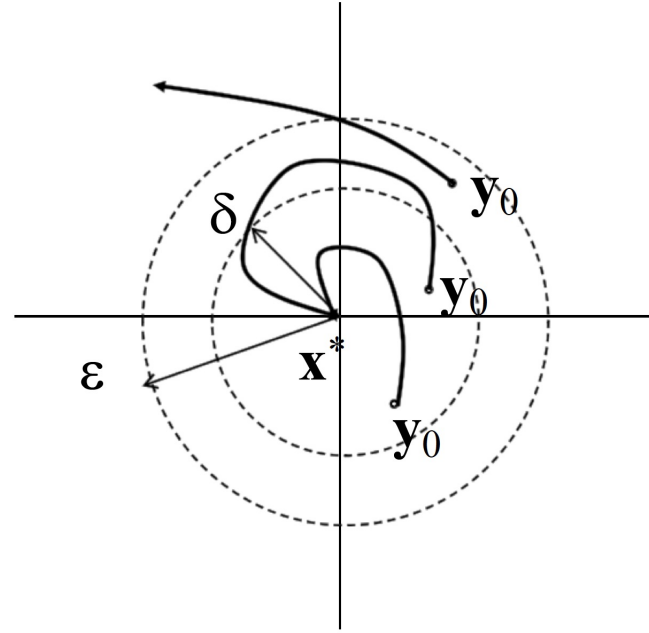
- **Definition:** An equilibrium point  $\mathbf{x}^*$  is asymptotically stable if it is stable and  $\beta > 0$  exists such that

$$\lim_{t \rightarrow \infty} \|\mathbf{x}(t) - \mathbf{x}^*\| = 0 \text{ whenever } \|\mathbf{x}(0) - \mathbf{x}^*\| < \beta$$

# Picturing stability



Stable equilibrium



Asymptotically stable equilibrium

- The solution cannot escape from a **stable** equilibrium
- The solution converges to the equilibrium point if it starts close enough to an **asymptotically stable** equilibrium

# Epsilon-delta arguments

- Observe that the argument made within a definition like

$$\|\mathbf{x}(t) - \mathbf{x}^*\| < \varepsilon \text{ for all } t \geq 0 \text{ whenever } \|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$$

takes the form of a game:

1. I give you a positive number  $\varepsilon$  that I am free to choose
  2. you respond with a number  $\delta$  that satisfies some condition
  3. if you can find a number  $\delta$  for any  $\varepsilon$ , you 'win'
- Many mathematical proofs and definitions are based on this kind of argument

# Exponential stability

- **Definition:** An equilibrium point  $\mathbf{x}^*$  is **exponentially stable** if  $\mathbf{x}^*$  is asymptotically stable and there exist finite constants  $\alpha, \beta, \delta > 0$  such that

$$\|\mathbf{x}(t) - \mathbf{x}^*\| < \alpha e^{-\beta t} \|\mathbf{x}(0) - \mathbf{x}^*\| \quad \forall t \geq 0 \quad \text{whenever} \quad \|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$$

- As well as requiring that the solution is stable and converges to the equilibrium point (asymptotic stability), this also quantifies the **rate of convergence**

i.e. how fast the solution flows to the equilibrium point

# Flows in 2nd order linear systems

- Ultimately we will study flows around the equilibrium points of nonlinear ODE systems by examining **local linearizations** around those points
- Each flow has a **topology** (a shape) that falls into one of a number of distinct categories
- The flows in local linearizations can often be continuously distorted into flows that solve the nonlinear ODE systems
- It is useful to study the topologies of some example linear systems to understand what families of solutions look like

# Uncoupled second order linear system

- Perhaps the simplest problem we can think of is

$$\begin{array}{ll} \frac{dx_1}{dt} = \alpha_1 x_1 & \\ \frac{dx_2}{dt} = \alpha_2 x_2 & \end{array} \quad \text{solved by} \quad \begin{array}{l} x_1(t) = x_1(0)e^{\alpha_1 t} \\ x_2(t) = x_2(0)e^{\alpha_2 t} \end{array}$$

- These can be viewed as parametric equations that describe the shapes of curves in phase space

$$\begin{aligned} (x_1(t))^{\alpha_2/\alpha_1} &= (x_1(0))^{\alpha_2/\alpha_1} e^{\alpha_2 t} = \frac{(x_1(0))^{\alpha_2/\alpha_1}}{x_2(0)} x_2(t) \\ \implies x_2 &= c x_1^{\alpha_2/\alpha_1} \end{aligned}$$

# Stability of the decoupled system

- Let's look more closely at the system

$$\begin{array}{ll} \frac{dx_1}{dt} = \alpha_1 x_1 & \\ \frac{dx_2}{dt} = \alpha_2 x_2 & \end{array} \quad \text{solved by} \quad \begin{array}{l} x_1(t) = x_1(0)e^{\alpha_1 t} \\ x_2(t) = x_2(0)e^{\alpha_2 t} \end{array}$$

- The system has an equilibrium point at the origin (because it's an autonomous linear system)
- If  $\alpha_1 < 0$  and  $\alpha_2 < 0$ , then the system is asymptotically stable, in fact **exponentially stable**
- If  $\alpha_1 > 0$  or  $\alpha_2 > 0$ , then the system is **unstable**

# Coupled second order linear system

- For a coupled linear system the analysis is slightly more complicated – consider

$$\begin{aligned}\frac{dx_1}{dt} &= ax_1 + bx_2 \\ \frac{dx_2}{dt} &= cx_1 + dx_2\end{aligned}\quad \text{or} \quad \dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \text{ where } \mathbf{A} = \begin{bmatrix} a & d \\ c & b \end{bmatrix}$$

- Here the solution for initial condition  $\mathbf{x}(0)$  is  $\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}(0)$
- To simplify the matrix exponential term, diagonalize matrix  $\mathbf{A}$  (assuming it is diagonalizable):

$$e^{t\mathbf{A}} = \mathbf{V}\text{diag}\{e^{\lambda_i t}\}\mathbf{V}^{-1}$$

and rewrite the solution in terms of eigenvalues and eigenvectors

# Eigenvalues

- Here the eigenvalues of  $\mathbf{A}$  are found by solving

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{bmatrix} a - \lambda & d \\ c & b - \lambda \end{bmatrix} = \lambda^2 - (a + b)\lambda + (ab - cd) = 0$$

- This characteristic equation can also be written in terms of the **trace** and **determinant** of  $\mathbf{A}$ :

$$\lambda^2 - \text{tr}(\mathbf{A})\lambda + \det(\mathbf{A}) = 0$$

Recall that the trace of a matrix is the sum of its eigenvalues:

$$\text{tr}(\mathbf{A}) = \lambda_1 + \lambda_2$$

and its determinant is the product of its eigenvalues:

$$\det(\mathbf{A}) = \lambda_1 \lambda_2$$

# Solving with eigenvectors and eigenvalues

- If the eigenvalues of  $\mathbf{A}$  are real and distinct, then we can write any initial condition as a combination of the eigenvectors:

$$\mathbf{x}(0) = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \mathbf{V} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \mathbf{V}^{-1} \mathbf{x}(0)$$

- Then, using  $\mathbf{x}(t) = e^{t\mathbf{A}} \mathbf{x}(0)$ , we get

$$\mathbf{x}(t) = e^{t\mathbf{A}} \mathbf{V} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \mathbf{V} \text{diag}\{e^{\lambda_i t}\} \mathbf{V}^{-1} \mathbf{V} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$$

- The solution is a linear combination of exponential transients with decay rates determined by the eigenvalues

# Characteristics of linear system trajectories

- If  $\text{Re}(\lambda) < 0$ , then the component of the solution along the corresponding eigenvector **decays to zero**
- If  $\text{Re}(\lambda) > 0$ , then the component of the solution along the corresponding eigenvector **grows exponentially**
- If  $\text{Re}(\lambda) = 0$ , then the component of the solution along the corresponding eigenvector **remains constant**
- If  $\text{Im}(\lambda) \neq 0$ , then the solution orbits or **spirals** around the origin
- If  $\text{Im}(\lambda) = 0$ , then the solution tends towards the eigenvector with the dominant eigenvalue

# Coordinate transformation to normal form

Characteristic trajectory shapes in 2-D phase space are found using a coordinate transformation that puts  $\mathbf{A}$  into a standard form:

- given  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ , define new coordinates  $\mathbf{y} = \mathbf{K}^{-1}\mathbf{x}$
- this transforms the equations of motion to  $\dot{\mathbf{y}} = \mathbf{K}\mathbf{A}\mathbf{K}^{-1}\mathbf{y}$
- if the eigenvalues of  $\mathbf{A}$  are distinct and real, let  $\mathbf{K} = \mathbf{V}$  then  $\mathbf{V}\mathbf{A}\mathbf{V}^{-1} = \mathbf{D}$

$$\dot{\mathbf{y}} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \mathbf{y}$$

- if the eigenvalues of  $\mathbf{A}$  are complex,  $\lambda = a \pm jb$ , then let  $\mathbf{K} = \mathbf{V}'$ :

$$\dot{\mathbf{y}} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \mathbf{y} \quad \text{where} \quad \mathbf{V}' = [\mathbf{w} \quad \mathbf{u}], \quad \text{and} \\ \mathbf{v} = \mathbf{u} \pm j\mathbf{w} \quad \text{are the eigenvectors}$$

The transformed matrix is called the **normal form** of  $\mathbf{A}$

# The special case of a defective matrix

- If the eigenvalues of  $\mathbf{A}$  are real but not distinct, (e.g.  $\lambda_1 = \lambda_2$ ), then  $\mathbf{A}$  may not have  $n$  linearly independent eigenvectors (i.e.  $\mathbf{A}$  is **defective**)

e.g.  $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$  has eigenvalue  $\lambda = 1$  (multiplicity 2), eigenvector  $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

- In this case we can compute a **generalised eigenvector**  $\mathbf{v}_2$  using

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_2 = \mathbf{v}_1 \quad \implies \quad (\mathbf{A} - \lambda_1 \mathbf{I})^2 \mathbf{v}_2 = (\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{v}_1 = 0$$

If  $\lambda_i$  has multiplicity  $m$ , it has  $m$  linearly independent generalised eigenvectors

- Defining the transformation matrix  $\mathbf{V}' = [\mathbf{v}_1 \quad \mathbf{v}_2]$  allows us to express  $\mathbf{A}$  as

$$\mathbf{A} = \mathbf{V}' \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} (\mathbf{V}')^{-1}$$

This is called the **Jordan normal form** of  $\mathbf{A}$

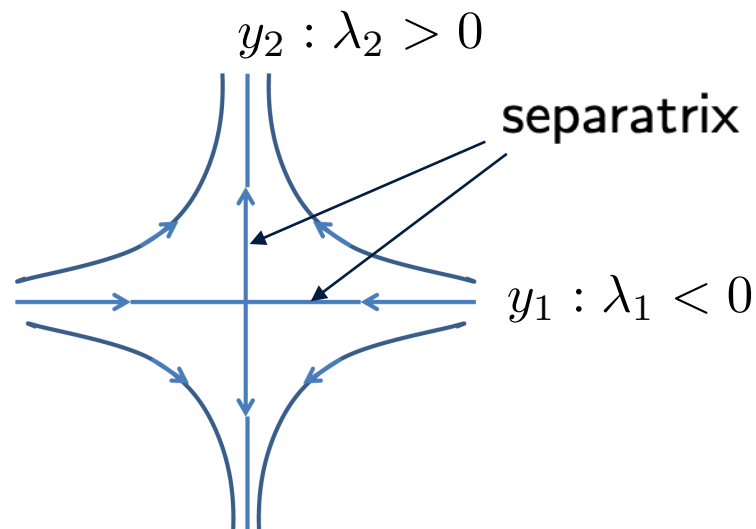
# Saddle equilibrium points via normal forms

- If the eigenvalues of  $\mathbf{A}$  are real and  $\lambda_1 \lambda_2 < 0$ , then the equilibrium  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  is unstable – this results in a **saddle point**

In transformed coordinates:  $\mathbf{V}^{-1}\mathbf{A}\mathbf{V} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

- The phase portrait has four asymptotes: two correspond to solutions that approach the origin as  $t \rightarrow \infty$ , and two as  $t \rightarrow -\infty$
- These four trajectories are called **separatrices**

a saddle shape in  
the phase plane:



# Stable equilibrium points via normal forms

- If the eigenvalues of  $\mathbf{A}$  are real and both  $\lambda_1 < 0$  and  $\lambda_2 < 0$ , then the equilibrium will be **stable**

- Three cases:

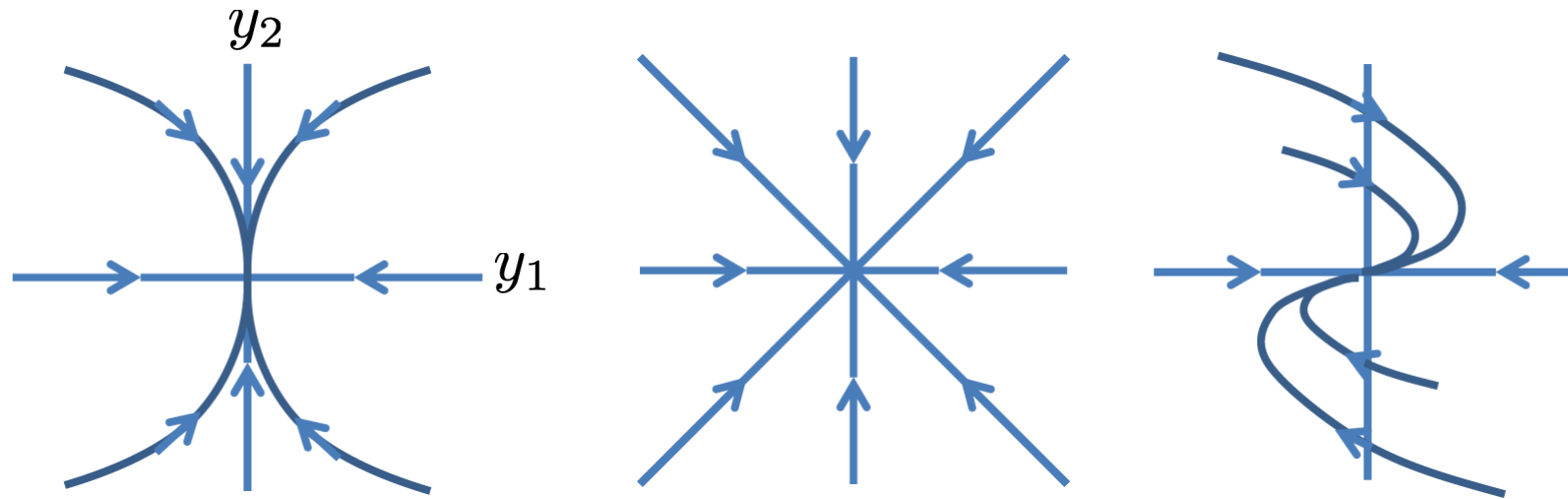
- Eigenvalues distinct:  $\mathbf{V}^{-1}\mathbf{A}\mathbf{V} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

- Eigenvalues repeated but two eigenvectors:  $\mathbf{V}^{-1}\mathbf{A}\mathbf{V} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{bmatrix}$

- Eigenvalues repeated but  $\mathbf{A}$  defective:  $(\mathbf{V}')^{-1}\mathbf{A}\mathbf{V}' = \begin{bmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{bmatrix}$

# Stable and unstable flow shapes

- Consider the transformed coordinates:  $\mathbf{y} = \mathbf{V}^{-1}\mathbf{x}$  or  $\mathbf{y} = (\mathbf{V}')^{-1}\mathbf{x}$
- Stable solutions when both eigenvalues are real and negative:



standard case:  $\lambda_1 < \lambda_2$     nondegenerate:  $\lambda_1 = \lambda_2$     degenerate

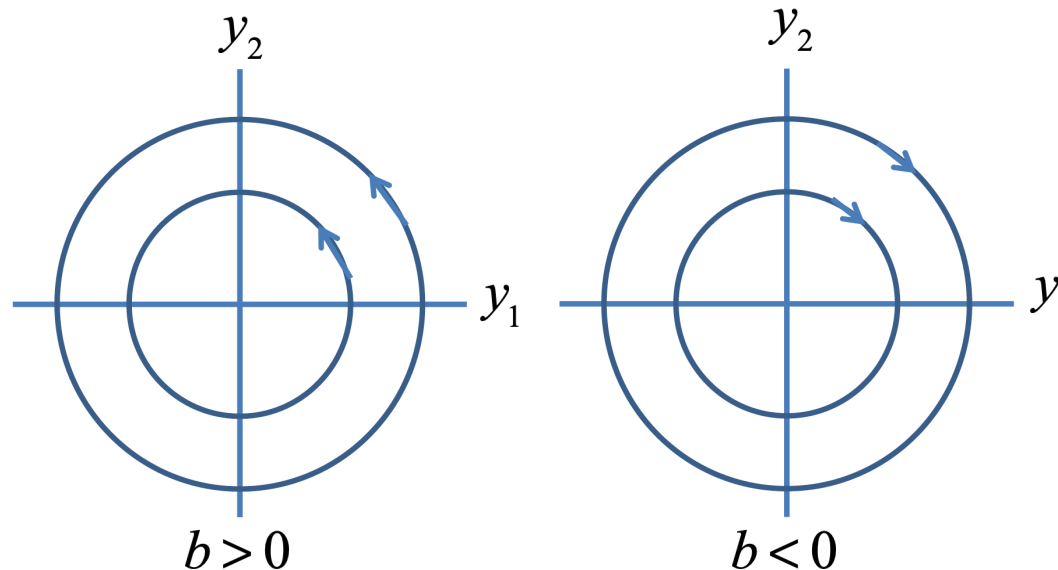
- If both eigenvalues are positive then the solutions are unstable; flows in the phase plane look the same as above, but the arrows point in the opposite directions

# Centre equilibrium points via normal forms

- If the eigenvalues are purely imaginary (real part equal to zero), then the corresponding equilibrium point is marginally stable
- In this case  $\lambda = 0 \pm jb$  and the transformation on slide 17 shows that the normal form is

$$(\mathbf{V}')^{-1} \mathbf{A} \mathbf{V}' = \begin{bmatrix} 0 & -b \\ b & 0 \end{bmatrix}$$

- Phase portraits:



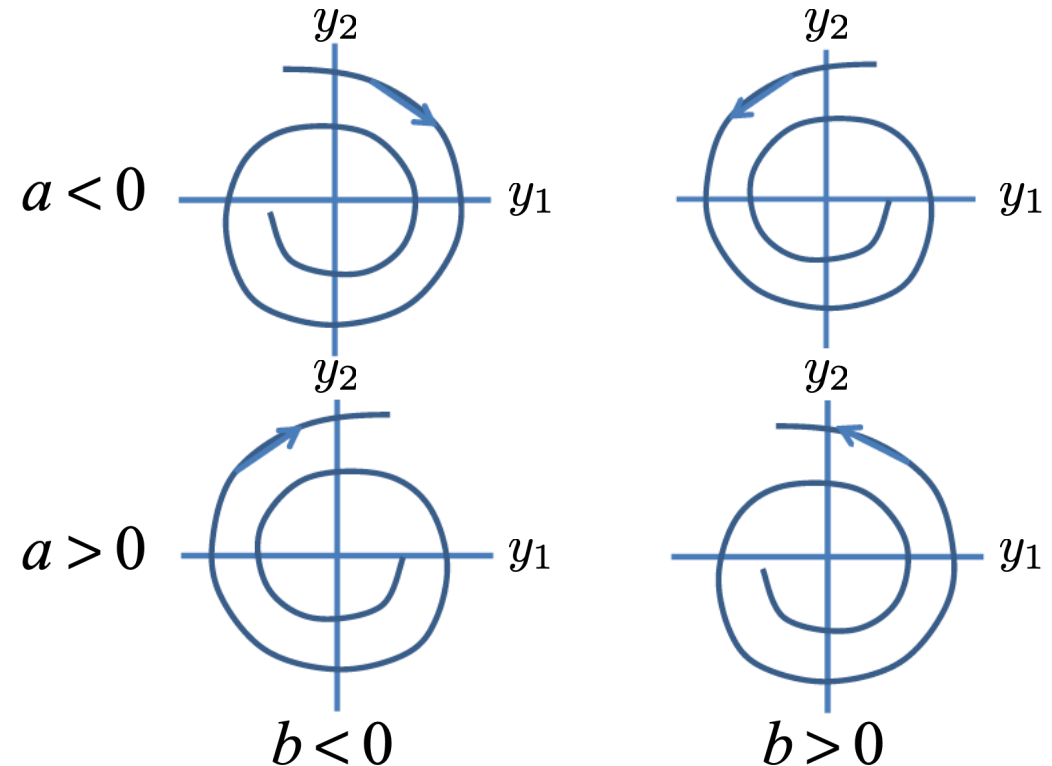
# Stable and unstable spirals

- If the eigenvalues are  $\lambda_1 = a + jb$ ,  $\lambda_2 = a - jb$ , then the normal form is

$$(\mathbf{V}')^{-1} \mathbf{A} \mathbf{V}' = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

- Phase portraits are **spirals**

$a < 0$  : spiral is stable  
 $a > 0$  : spiral is unstable  
 $b > 0$  : spiral is clockwise  
 $b < 0$  : spiral is anticlockwise



# Phase space equation summary

- We have sketched how solutions behave in the phase plane using the normal forms of the system  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  with  $\mathbf{x}(0) = \mathbf{x}_0$ :

$$\dot{\mathbf{y}} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \mathbf{y} \implies \mathbf{y}(t) = \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} \mathbf{y}_0 \quad \text{real; complete eigenvector set}$$

$$\dot{\mathbf{y}} = \begin{bmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{bmatrix} \mathbf{y} \implies \mathbf{y}(t) = e^{\lambda_1 t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \mathbf{y}_0 \quad \text{real; degenerate}$$

$$\dot{\mathbf{y}} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \mathbf{y} \implies \mathbf{y}(t) = e^{at} \begin{bmatrix} \cos bt & -\sin bt \\ \sin bt & \cos bt \end{bmatrix} \mathbf{y}_0 \quad \text{complex}$$

## Example of coordinate change

- Consider the system  $\dot{\mathbf{x}} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix} \mathbf{x}$
- Eigenvalues from characteristic equation:  $\lambda^2 + \lambda - 2 = 0 \implies \lambda \in \{1, -2\}$
- Eigenvectors:  $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ ,  $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$   
$$\implies \mathbf{A} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}^{-1}$$
- Define new coordinates  $\mathbf{y} = \mathbf{V}^{-1} \mathbf{x}$ , then the transformed coordinate axes are:

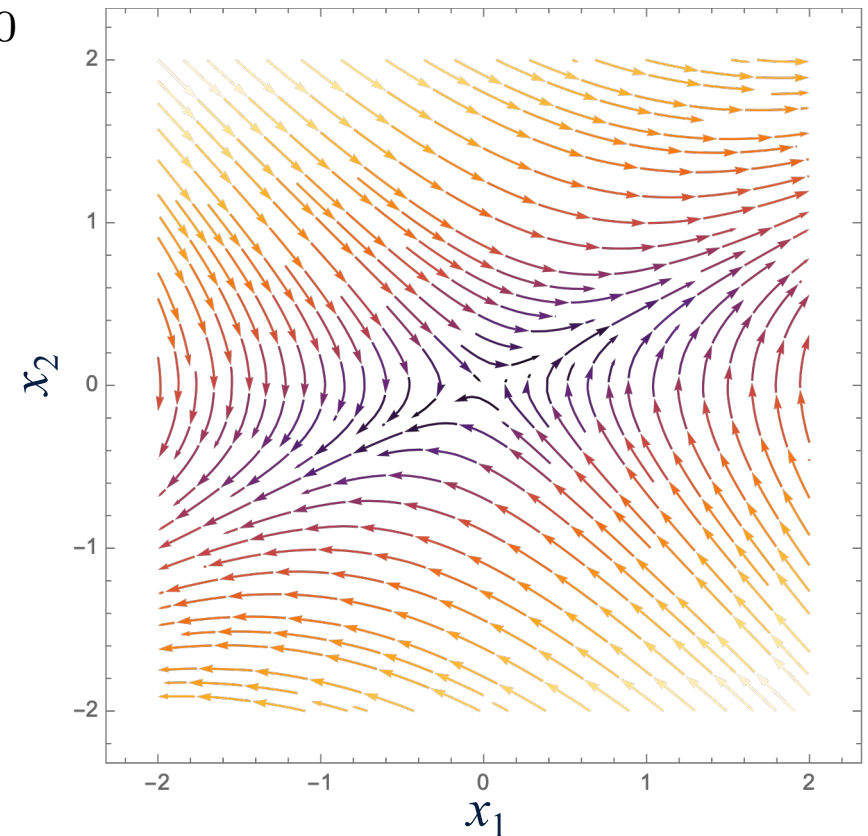
$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ (unstable), and } \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ (stable)}$$

## Example continued

- Transformed system:  $\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}^{-1} \dot{\mathbf{x}} = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}^{-1} \mathbf{x}$

- Solution:  $\mathbf{x}(t) = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}^{-1} \mathbf{x}_0$

$$\mathbf{x}(t) = \frac{1}{3} \begin{bmatrix} e^{-2t} + 2e^t & -2e^{-2t} + 2e^t \\ -e^{-2t} + e^t & 2e^{-2t} + e^t \end{bmatrix} \mathbf{x}_0$$



# Linearization of nonlinear systems

- Consider the nonlinear system  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$  with equilibrium at  $\mathbf{x}^*$
- Let  $\mathbf{x} = \mathbf{x}^* + \mathbf{w}$  and assume that  $\mathbf{f}$  is differentiable.  
Then the Taylor expansion of the  $i$ th entry,  $f_i$ , of  $\mathbf{f}$  gives

$$f_i(\mathbf{x}^* + \mathbf{w}) = f_i(\mathbf{x}^*) + \sum_{j=1}^n \left( \frac{\partial f_i}{\partial x_j} \Big|_{\mathbf{x}^*} w_j + O(|w_j|^2) \right)$$

- Noting that equilibrium is independent of time by definition

$$\dot{\mathbf{x}} = \dot{\mathbf{w}} = D\mathbf{f}(\mathbf{x}^*)\mathbf{w} + O(\|\mathbf{w}\|^2)$$

- The matrix  $D\mathbf{f}(\mathbf{x})$  is called the **Jacobian** of the vector-valued function  $\mathbf{f}$ , with  $ij$ th entry  $(D\mathbf{f})_{ij} = \partial f_i / \partial x_j$

# Hyperbolic equilibria

**Definition** (hyperbolic equilibrium): If  $\mathbf{x}^*$  is an equilibrium of system  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ , then  $\mathbf{x}^*$  is called a *hyperbolic fixed point* if all eigenvalues of the Jacobian  $D\mathbf{f}(\mathbf{x}^*)$  have nonzero real parts

- This leads to an important **theorem**: if an equilibrium point is a hyperbolic fixed point and all the eigenvalues of the Jacobian have negative real parts, then the equilibrium solution  $\mathbf{x} = \mathbf{x}^*$  is asymptotically stable
- Note that this proves asymptotic stability but does not say anything about the size of the region of stability
  - this depends on the size of  $\delta$  from the game on slide 8
- In lecture 3 we will see how linearizations near equilibrium points can be used to get information about nonlinear systems

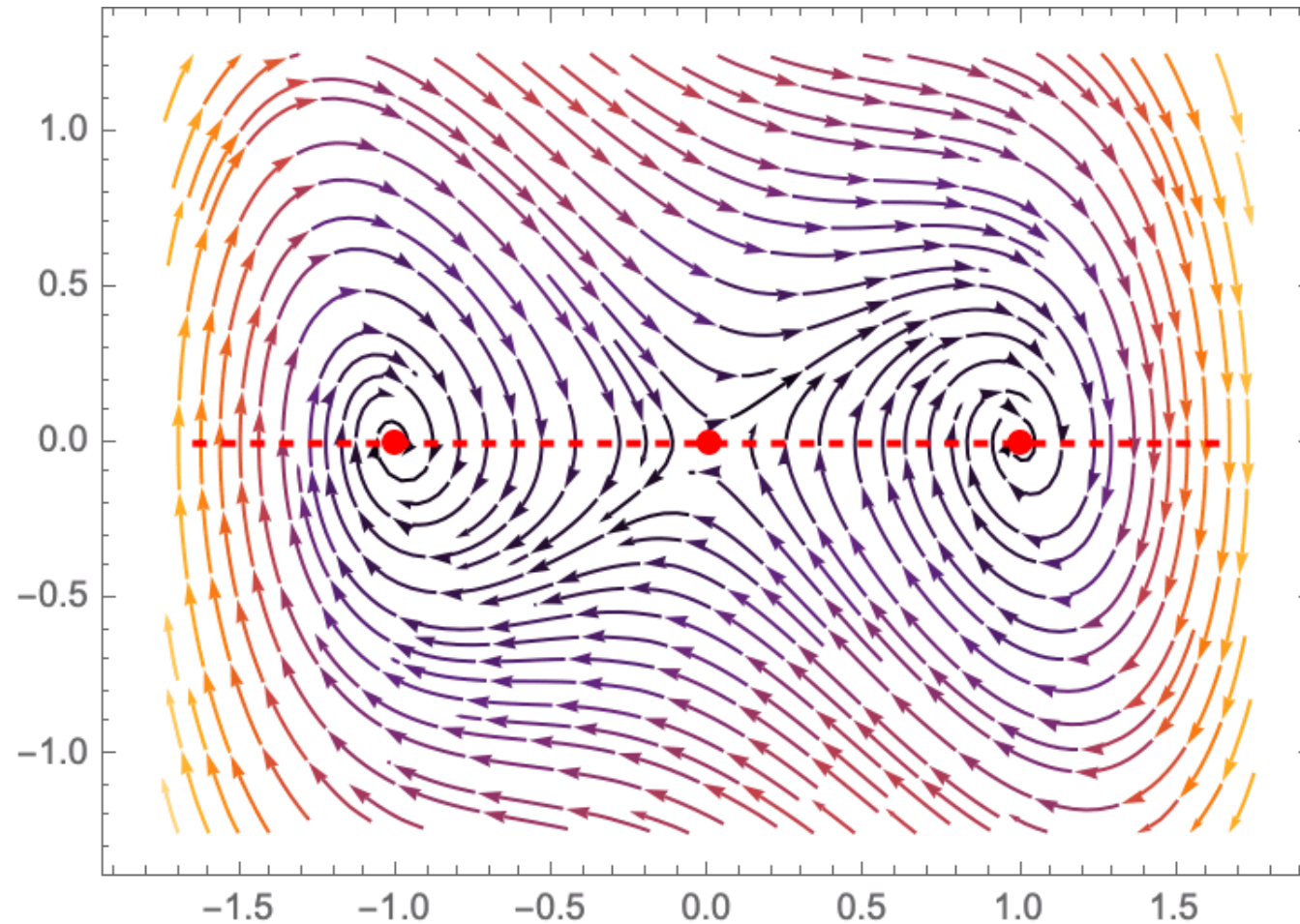
## Example: Duffing oscillator

The Duffing oscillator is described for  $\gamma \geq 0$  by

$$\begin{aligned}\frac{dx}{dt} &= y \\ \frac{dy}{dt} &= x - x^3 - \gamma y\end{aligned}$$

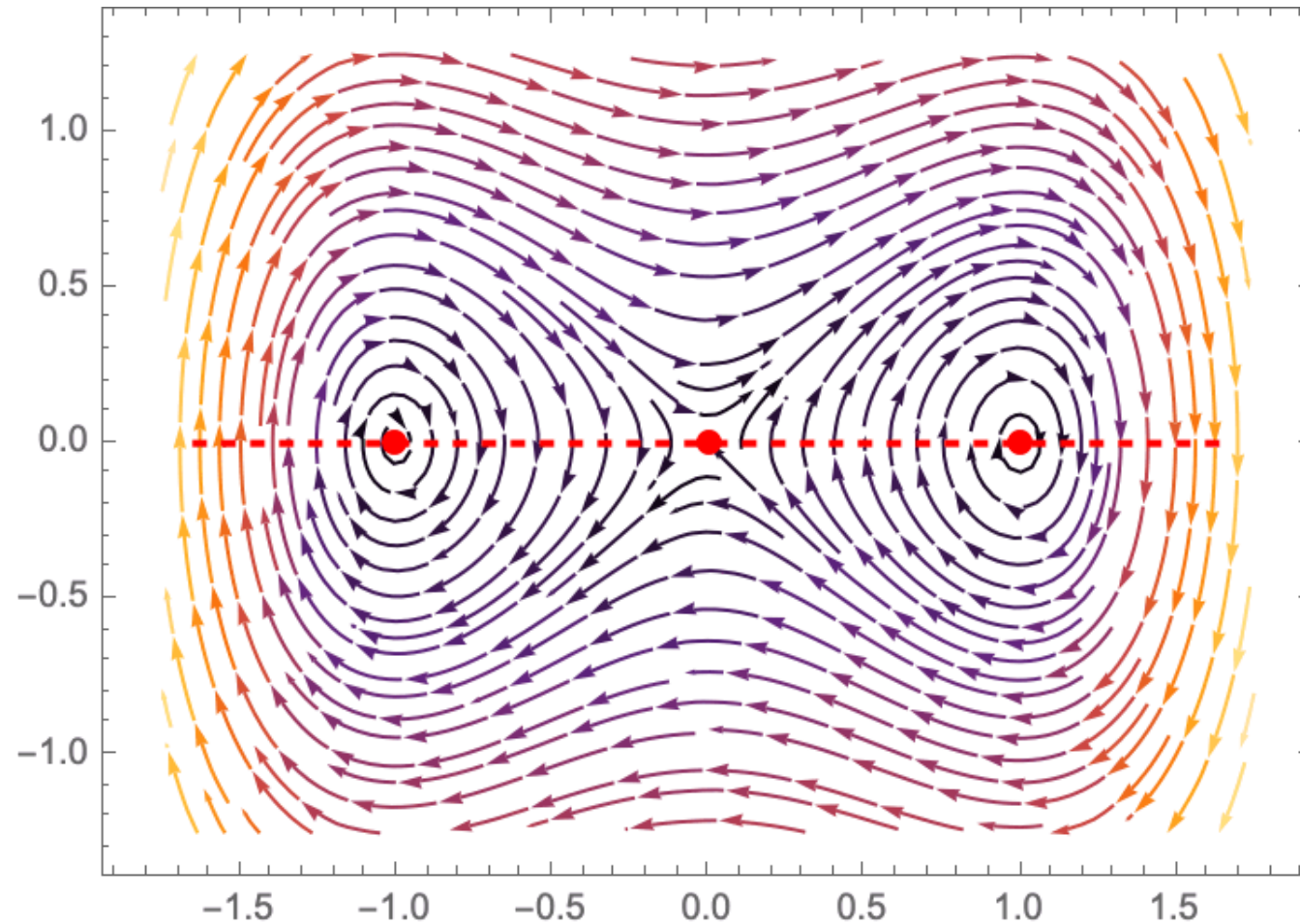
- equilibria:  $(x^*, y^*) = (0, 0)$  and  $(\pm 1, 0)$
- Jacobian:  $D\mathbf{f} = \begin{bmatrix} 0 & 1 \\ 1 - 3x^2 & -\gamma \end{bmatrix}$
- at  $(0, 0)$  :  $\lambda_{1,2} = \frac{-\gamma \pm \sqrt{\gamma^2 + 4}}{2} \implies$  unstable
- at  $(\pm 1, 0)$  :  $\lambda_{1,2} = \frac{-\gamma \pm \sqrt{\gamma^2 - 8}}{2} \implies$  asymptotically stable if  $\gamma > 0$
- If  $\gamma = 0$ ,  $(\pm 1, 0)$  is a centre  $\implies$  local linearization inconclusive

## Example: Duffing oscillator



Phase plane of the Duffing oscillator with  $\gamma = 0.5$

## Example: Duffing oscillator



Phase plane of the Duffing oscillator with  $\gamma = 0$

# Questions?